CCNA 200-301 V2.0

18 OSPFv3 for IPv6

Part III — IP Routing

EXAM TOPICS

3.3

LECTURER

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LEARNING OUTCOMES

What you will be able to do

- **State** what OSPFv3 keeps from OSPFv2 and what it changes.
- **Configure** single-area OSPFv3 on interfaces rather than with network statements.
- **Explain** why an IPv6-only router still needs an IPv4-format router ID, and configure one.
- **Troubleshoot** an OSPFv3 adjacency, including the failure unique to it: a missing router ID.

ASSUMED KNOWLEDGE

Before we start

Chapter 17 — this chapter assumes it and covers only the differences. Chapter 5 for link-local addressing.

KEY TERMS

Vocabulary for this chapter

- OSPFv3
- router ID
- link-local next hop
- per-interface configuration
- address family
- ff02::5
- ff02::6

Each is defined where it first matters — not here.

ON THE EXAM

One topic, two protocols

This is what the blueprint actually asks for

Topic **3.3** covers OSPFv2 and OSPFv3 together, and OSPFv3 is new in v2.0 — it was not in the v1.1 blueprint at all. The exam will not ask you to re-learn OSPF; it will ask whether you can configure the v6 version, which means knowing the handful of things that differ.

WATCH OUT

The failure unique to OSPFv3

The router ID is still a 32-bit value written like an IPv4 address, even in a network with no IPv4 at all. If the router has *no* IPv4 address on any interface and you have not configured `router-id`, the process will not start. IOS says so, but the message is easily missed in a long paste:

% OSPFv3-1-IPV6 Router process 10 could not pick a router-id, please configure manually

Always configure it explicitly. It costs one line and removes the whole class of problem.

Single-area ospfv3 on r1

Configure

```
ipv6 unicast-routing
!
! The process exists mainly to hold the router ID; interfaces do
! the rest of the work.
ipv6 router ospf 10
 router-id 1.1.1.1
 passive-interface Vlan10
!
! Enable per interface. Every IPv6 prefix on the interface is
! advertised -- there is no network statement and no wildcard mask.
interface GigabitEthernet0/1
 description Link to R2
 ipv6 address 2001:db8:acad:99::1/64
 ipv6 ospf 10 area 0
!
interface Vlan10
 ipv6 address 2001:db8:acad:10::1/64
 ipv6 ospf 10 area 0
!
```

Single-area ospfv3 on r1 (continued)

Configure

```
! Point-to-point on a two-router link, exactly as in OSPFv2.
interface GigabitEthernet0/2
 ipv6 ospf network point-to-point
```

CONCEPT

No network statements is a simplification, not a trap

In OSPFv2 you had to compute a wildcard mask and remember that the statement both enables the interface and advertises the subnet. OSPFv3 drops all of that: enable the interface and every prefix on it is advertised, including additional addresses added later.

The practical consequence is that `passive-interface` becomes more important, not less — it is now the only way to advertise a LAN's prefix without sending hellos into it.

show ipv6 ospf neighbor

R1# show ipv6 ospf neighbor

```
              OSPFv3 Router with ID (1.1.1.1) (Process ID 10)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
2.2.2.2           1   FULL/BDR        00:00:31    4               GigabitEthernet0/1
3.3.3.3           1   FULL/DR         00:00:35    4               GigabitEthernet0/1
```

VERIFY

show ipv6 route ospf

R1# show ipv6 route ospf

```
IPv6 Routing Table - default - 9 entries
Codes: C - Connected, L - Local, S - Static, O - OSPF intra, OI - OSPF inter
O   2001:DB8:ACAD:20::/64 [110/2]
     via FE80::2, GigabitEthernet0/1
O   2001:DB8:ACAD:30::/64 [110/3]
     via FE80::3, GigabitEthernet0/1
```

WATCH OUT

The next hop is a link-local address, and that is correct

Every OSPFv3 route resolves to a `fe80::` next hop plus an interface. This looks wrong to anyone used to IPv4 and is not: link-local addresses are guaranteed to exist on every IPv6 interface and cannot be renumbered, so they are the stable choice. This is also why the interface must always be named alongside them (Chapter 16).

VERIFY

show ipv6 ospf interface brief

```
R1#  show ipv6 ospf interface brief

Interface      PID    Area               Intf ID     Cost   State Nbrs F/C
Gi0/1          10     0                  4           1      BDR   2/2
Gi0/2          10     0                  5           1      P2P   1/1
Vl10           10     0                  8           1      DR    0/0
```

Verifying

- **No router ID on an IPv6-only router.** The process will not start. This is the single most common OSPFv3 fault.
- **Looking for network statements.** There are none; enable the interface.
- **Forgetting `ipv6 unicast-routing`.** As ever. Interfaces come up, OSPFv3 forms adjacencies, and nothing is forwarded.
- **Expecting a global next hop in the routing table.** Link-local is correct.
- **Assuming OSPFv2 and OSPFv3 share anything at runtime.** They are separate processes with separate databases and separate neighbours. A working v2 adjacency tells you nothing about v3 on the same link.

OSPFv3 configured on both routers; no neighbours at all.

`R1` and `R2` both have IPv6 addresses on the link, ping each other's link-local addresses successfully, and have identical-looking OSPFv3 configuration. `show ipv6 ospf neighbor` is empty on both.

OSPFv3 configured on both routers; no neighbours at all.

What would you check first — and what do you expect to see?

show ipv6 ospf

R1# show ipv6 ospf

```
%OSPFv3: Router process 10 is not running, please configure a router-id
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. Neither router has any IPv4 address anywhere — this is an IPv6-only lab — so neither could derive a 32-bit router ID, and the process never started. The interface commands were accepted and are sitting in the configuration doing nothing, which is why the configuration looks right.

The tell is that *both* ends are empty. A mismatch of area, timers or MTU usually leaves at least one side reporting something.

starting the process

Configure

```
ipv6 router ospf 10
 router-id 1.1.1.1
```

And the lesson

Fix. Configure a router ID on each router explicitly:

The process starts immediately and adjacencies form within a hello interval. Configure a distinct ID on `R2` — duplicate router IDs are the next fault along, and they produce a flapping adjacency rather than none.

Dual-stack routing, both protocols side by side

Topology. The three-router topology from Chapter 17, now addressed for IPv4 and IPv6 simultaneously.

1. Keep the working OSPFv2 configuration. Add IPv6 addresses to every link and LAN, and `ipv6 unicast-routing` everywhere.
2. Configure OSPFv3 with explicit router IDs matching the v2 ones. Enable it per interface, LANs passive.
3. Confirm `show ipv6 ospf neighbor` reaches FULL on every link, and that the DR and BDR are the same routers as for OSPFv2. Explain why they need not have been.
4. Compare `show ip route ospf` with `show ipv6 route ospf`. Note that v4 next hops are global and v6 next hops are link-local.

Dual-stack routing, both protocols side by side (continued)

1. Remove the router ID from one router and clear the process. Reproduce the “process not running” fault, then fix it.
2. Set one link to `ipv6 ospf network point-to-point` at one end only and observe the result.
3. **Extension.** Shut a link and time how long each protocol takes to reconverge. They should be comparable — the algorithm is the same.

CHECKPOINT 1 of 3

Why does an IPv6-only router still need a router ID in IPv4 format, and what happens if it cannot get one?

Discuss · then advance

ANSWER

Checkpoint 1 of 3

Why does an IPv6-only router still need a router ID in IPv4 format, and what happens if it cannot get one?

> Because the router ID is a 32-bit value written in IPv4 format regardless of the address family, and it is normally derived from an IPv4 address on the device. With no IPv4 anywhere, none can be derived, and the process will not start at all — the interface commands are accepted and do nothing.

CHECKPOINT 2 of 3

OSPFv2 works on a link and OSPFv3 does not. What does the working v2 adjacency prove about the v3 problem?

Discuss · then advance

ANSWER

Checkpoint 2 of 3

OSPFv2 works on a link and OSPFv3 does not. What does the working v2 adjacency prove about the v3 problem?

Nothing. They are separate processes with separate databases, neighbours and adjacencies. A working v2 adjacency proves the link and layer 2 are fine, which is useful, but says nothing about v3's configuration.

CHECKPOINT 3 of 3

Where is OSPFv3 enabled, and what is advertised as a result?

Discuss · then advance

ANSWER

Checkpoint 3 of 3

Where is OSPFv3 enabled, and what is advertised as a result?

> Per interface, with `ipv6 ospf <pid> area <n>`. *Every* IPv6 prefix on that interface is advertised — there are no network statements and no wildcard masks, which is why `passive-interface` matters more in OSPFv3 than in OSPFv2.

What to take away

- OSPFv3 is OSPF. The database, the algorithm, the states, the DR election and the troubleshooting are unchanged.
- It is enabled per interface with `ipv6 ospf <pid> area <n>`, and every prefix on that interface is advertised. No network statements, no wildcards.
- Neighbours are reached by link-local address, so routes show a `fe80::` next hop plus an interface. That is correct.
- The router ID is still 32-bit IPv4 format and **must** be configured on an IPv6-only router, or the process will not start.
- v2 and v3 are independent processes. One working says nothing about the other.

Where we got to

- OSPFv3 is OSPF. The database, the algorithm, the states, the DR election and the troubleshooting are unchanged.
- It is enabled per interface with `ipv6 ospf <pid> area <n>`, and every prefix on that interface is advertised. No network statements, no wildcards.
- Neighbours are reached by link-local address, so routes show a `fe80::` next hop plus an interface. That is correct.
- The router ID is still 32-bit IPv4 format and **must** be configured on an IPv6-only router, or the process will not start.

NEXT

Chapter 19 — First Hop Redundancy: HSRP and VRRP